TEK4090

Introduction to Modern Control

Lecture 6: Feedforward Control

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Recap from Last Time

- Observability

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- Observability
- Estimators & State Estimation

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- Optimal State Estimation: Kalman Filtering

Recap from Last Time

- Observability
- Estimators & State Estimation
- Optimal State Estimation: Kalman Filtering
- Applications to Spacecraft GNC

$\tilde{y} = Cx + v$ $v \sim N(0, Q)$

reconstruct $\hat{x} \sim N(x, P)$

$u = -K\hat{x} + k_r r$

what we predict

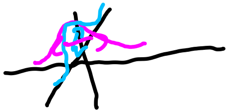
measure

$\hat{x}_k^+ = \hat{x}_k^- + K_k(\tilde{y} - C\hat{x}_k^-)$

un-updated state

Kalman gain

$K_k = \operatorname{argmin} \operatorname{tr}(P_k^+)$



Reading Assignment

System Inversion

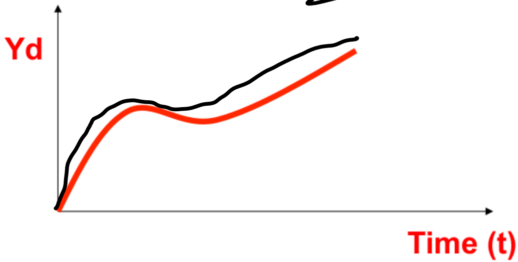
1. Notes by Santosh Devasia (University of Washington)

Motivation

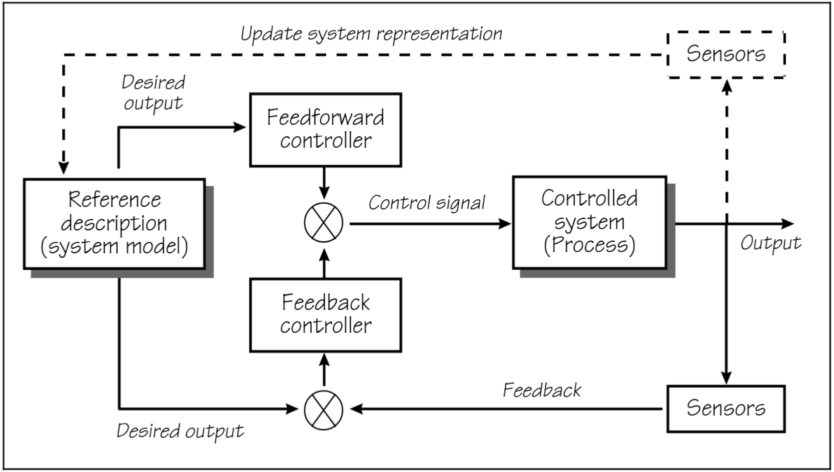
$$G(s)\begin{cases}\dot{x} = Ax + Bu\\ y = Cx\end{cases}$$

$$u \longrightarrow \boxed{G(s)} \longrightarrow y$$

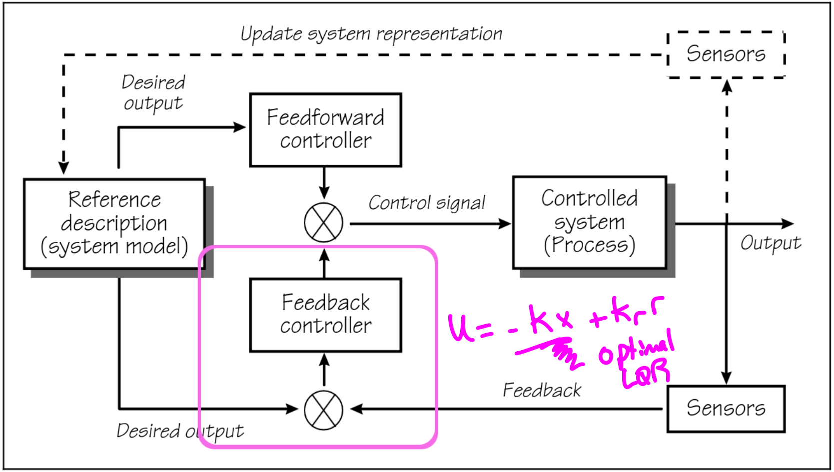
Goal: Find the input $u(t)$ that achieves a desired output trajectory $y(t)$



Control Block Diagram



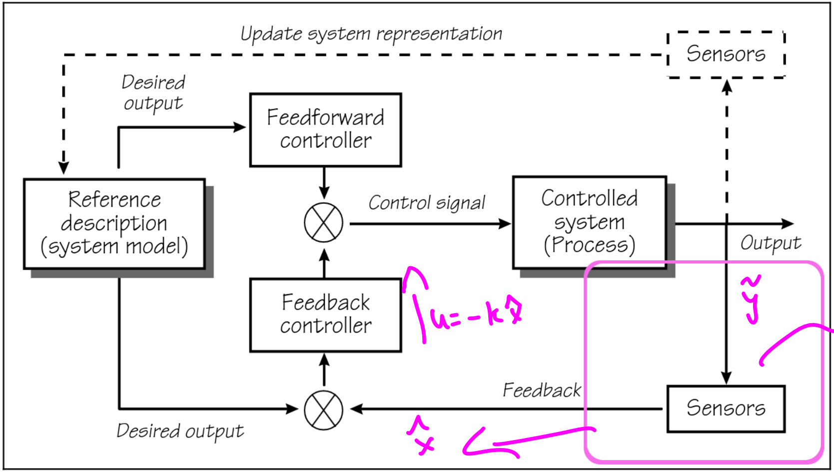
Control Block Diagram – Feedback



$u = -Kx + k_r r$

optimal LQR

Control Block Diagram – Estimation



$u = -K\hat{x}$

$\tilde{y}$

$\hat{x}$

$\hat{x}$ from $\tilde{y} = Cx + v$

Control Block Diagram – System Identification

First-principles (physics)

$F = ma = u$

MPC → DeePC → Lec 8

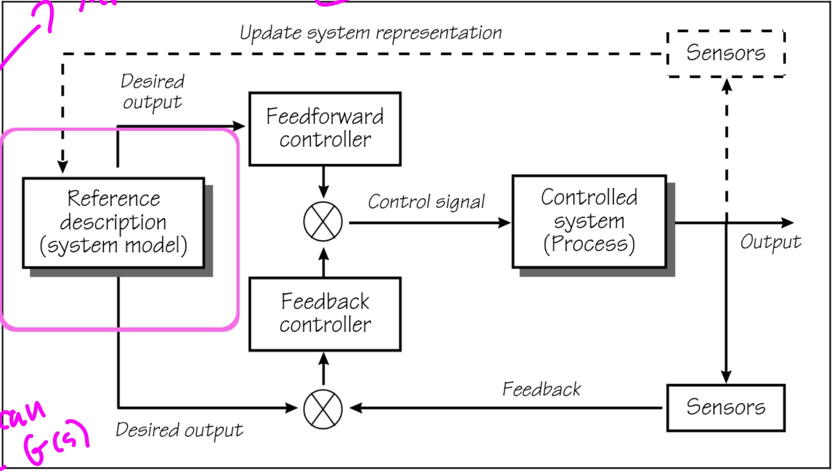
Data →

System identification

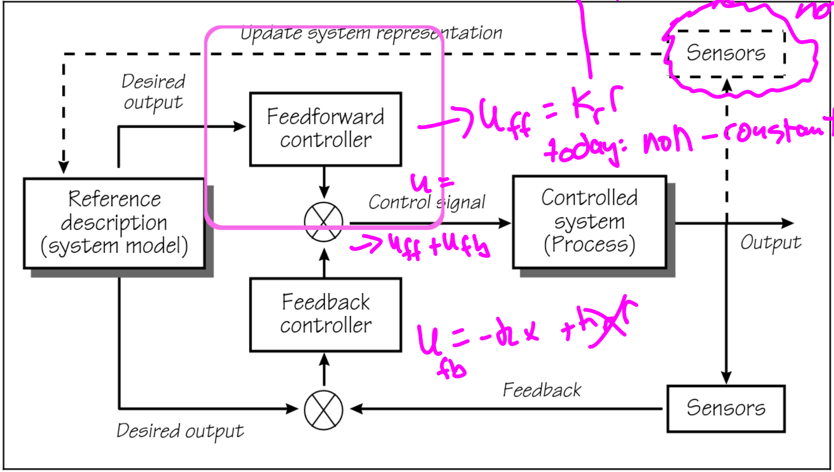
↓

design u to get y

so that you can estimate $G(s)$



Control Block Diagram – Feedforward Control



constant ref

not today

$\to u_{ff} = K_r r$

today: non-constant

$u =$

$\to u_{ff} + u_{fb}$

$u_{fb} = -Kx + \cancel{K_r r}$

System Inversion

$$\underbrace{Y(s)}_{\text{calculate}} = G(s)\underbrace{U(s)}_{\text{given}}$$

given $Y(s)$ $\{y(t)\}$
how do we get $U(s)$ $\{u(t)\}$?

Basic Example: Minimum-Phase SISO system

Transfer Function

$$\frac{Y(s)}{U(s)} := G(s) = \frac{(s+1)}{(s+2)}$$

$$\implies Y(s) = \frac{(s+2)}{(s+1)}U(s)$$

System Inversion

$$U(s) = G^{-1}(s)\, Y(s)$$

Basic Example: Minimum-Phase SISO system

reciprocal

Transfer Function

$$\frac{Y(s)}{U(s)} := G(s) = \frac{(s+1)}{(s+2)}$$

$$\implies Y(s) = \frac{(s+2)}{(s+1)} U(s)$$

Inverse Transfer Function

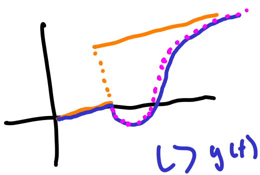
$$\frac{U(s)}{Y(s)} := G^{-1}(s) = \frac{(s+1)}{(s+2)}$$

$$\implies U(s) = \frac{(s+2)}{(s+1)} Y(s)$$

System Inversion

nonminimum phase → zeros of $G(s)$ LHP

↳ poles of $G^{-1}(s)$ → LHP



— ref
— sys (nmph)
⋯ yd

↳ $y(t)$ matches y_d

Basic Example: Minimum-Phase SISO system

↳ clearly, the "inverse" mapping works, in that I have $u(t)$ that gives me $y(t)$

Transfer Function

$$\frac{Y(s)}{U(s)} := G(s) = \frac{(s+1)}{(s+2)} \to \text{RHS}$$

$$\implies Y(s) = \frac{(s+2)}{(s+1)} U(s)$$

Inverse Transfer Function

$$\frac{U(s)}{Y(s)} := G^{-1}(s) = \frac{(s+2)}{(s+1)} \to \text{RHS}$$

$$\implies \underbrace{U(s) = \frac{(s+2)}{(s+1)} Y(s)}_{\text{"unstable"}}$$

Note that both $G(s)$ and $G^{-1}(s)$ are **open-loop stable**

Setup

single input / output

Linear SISO System

$$\dot{x} = Ax + Bu$$
$$y = Cx$$

$u, y \in \mathbb{R}$ C, B are vectors

Transfer Function

$$G(s) = C(sI - A)^{-1}B$$

$\hookrightarrow \mathbb{C}$

Setup

"weaker observability" condition

SISO

Assumption: Well-defined relative degree.

The system has a well-defined relative degree r if,

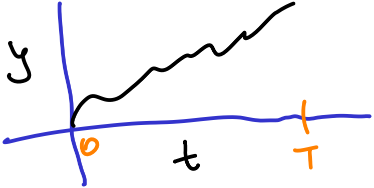
$$
\begin{aligned}
CA^j B &= 0 \qquad j < r - 1 \\
\underbrace{CA^{r-1} B}_{} &\neq 0
\end{aligned}
$$

Invertible!

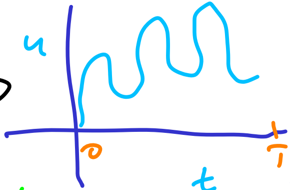
Setup

The relative degree r corresponds to the difference between the order of the denominator and the numerator of the transfer function

The Plan



y_d $\longleftrightarrow$ u_{ref}



$$\dot{x} = Ax + b\,u_{ref}$$

$$\hookrightarrow y = Cx = y_d(t)$$

"inverse system" $\longleftrightarrow$

"internal state"

$$\dot{\eta} = A_{inv}\boxed{\eta} + B_{inv}\boxed{y_d}$$

desired trajectory

$$\boxed{u} = C_{inv}\,\eta + D_{inv}\,y_d$$

$\hookrightarrow$ output $= u_{ref}$

Step 1: Differentiate Until the Output Appears

$$y = Cx(t)$$

$$\dot{y} = C\overbrace{(Ax + Bu)}^{\dot{x}} = CAx + \underbrace{CBu}_{\to\, CA^0Bu \,\to\, =0}$$

$$\ddot{y} = CA\dot{x} = CA(Ax + Bu) = CA^2x + \cancel{CABu}^{\,0}$$

$$\vdots$$

$$\frac{d^r}{dt^r}y := y^{(r)} = CA^rx + \underbrace{CA^{r-1}Bu(t)}_{\to\, \neq 0}$$

$$\hookrightarrow \quad y^{(r)} = CA^rx + CA^{r-1}Bu$$

$CA^jB = 0$ if $j < r-1$

$\downarrow$

$CA^{r-1}B \neq 0$

Step 1: Differentiate Until the Output Appears

invertible!

step 1.5: define the inverse input

$$y^{(r)}(t) = \underbrace{CA^{r}}_{A_y} x(t) + \underbrace{CA^{r-1}B}_{B_y \,\neq 0} u(t) = A_y x(t) + B_y u(t)$$

$$u_{inv}(t) = B_y^{-1}\left(y^{(r)}(t) - A_y \underbrace{x(t)}_{\text{unknown (partially)}} \right)$$

Problem: I know $y(t)$

I don't know $x(t)$

I'm not assuming observability

so, I can't (easily) get $x(t)$ from $y(t)$

Step 2: Decouple the State into Known/Unknown Parts

observability lecture: $\begin{bmatrix} y \\ \dot{y} \\ \vdots \\ y^{n-1} \end{bmatrix} = \begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix} x$

full col. rank, then we get x from y (and derivatives)

"known"

$$z(t) = \left[\begin{array}{c} y_d(t) \\ \dot{y}_d(t) \\ \vdots \\ y^{(r-1)}(t) \\ \hline \eta(t) \end{array}\right] := \left[\begin{array}{c} \mathcal{Y}_d(t) \\ \hline \eta(t) \end{array}\right] = \left[\begin{array}{c} C \\ CA \\ \vdots \\ CA^{r-1} \\ \hline T_\eta \end{array}\right] x = \underbrace{\left[\begin{array}{c} T_{\mathcal{Y}} \\ \hline T_\eta \end{array}\right]}_{T} x(t)$$

"unknown"

T_η is defined thusly: arbitrarily, so T is invertible

Step 2: Decouple the State into Known/Unknown Parts

$$x_{ref} = T^{-1}\begin{bmatrix} y_d(t) \\ \eta_{ref}(t) \end{bmatrix}$$

$\to \eta(t)$, so that $x(t)$ does what it needs to do so $y = Cx = y_d$

$\hookrightarrow$ need to figure out how to compute

$$= \left[\, T_l^{-1} \mid T_r^{-1} \,\right]\begin{bmatrix} y_d(t) \\ \eta_{ref}(t) \end{bmatrix}$$

notational abuse

$$= T_l^{-1} y_d(t) + T_r^{-1}\eta_{ref}(t)$$

$\to$ plug this back into $\dot{u}_{inv}$ to get a system of eqns for η_{ref}, which we can then solve

Step 3: Write the Inverse Input Dynamics

$u = C_{inv}\eta + D_{inv}\mathcal{Y}_d$

$$u_{inv} = B_y^{-1}\left(y_d^{(r)} - A_y x_{ref}\right)$$

$$= B_y^{-1}\left(y_d^{(r)} - A_y\left(T_e^{-1}\mathcal{Y}_d + T_r^{-1}\eta_{ref}\right)\right)$$

$$= B_y^{-1}y_d^{(r)} - B_y^{-1}A_y T_e^{-1}\mathcal{Y}_d - B_y^{-1}A_y T_r^{-1}\eta_{ref}$$

known

internal dynamics, unknown

stack)

$$= \underbrace{-B_y^{-1}A_y T_r^{-1}\eta_{ref}}_{C_{inv}\eta_{ref}} + \underbrace{\left[-B_y^{-1}A_y T_e^{-1} \;\middle|\; B_y^{-1}\right]}_{D_{inv}} \underbrace{\begin{bmatrix}\mathcal{Y}_d \\ \hline y_d^{(r)}\end{bmatrix}}_{\mathcal{Y}_d}$$

Step 3: Write the Inverse Input Dynamics

$u_{inv} = C_{inv}\,\eta(t) + D_{inv}\,y_d$

↳ we don't know this yet!

short & fat matrix

$z = \begin{bmatrix} y_d \\ \hline \eta \end{bmatrix} = T x$

↳ $z\begin{bmatrix} T_y \\ \hline T_\eta \end{bmatrix} x_z$

$\eta = \overbrace{T_\eta}^{} x$

recover η dynamics from x!

Step 3: Write the Inverse Input Dynamics

$$\eta = T_\eta x \longrightarrow \text{chosen arbitrarily to make } T \text{ invertible}$$

$$x_{\cancel{ref}} = T^{-1}\begin{bmatrix} y_d \\ \eta_{\cancel{ref}} \end{bmatrix} = \left[T_l^{-1} \mid T_r^{-1} \right] \begin{bmatrix} y_d \\ \eta_{\cancel{ref}} \end{bmatrix}$$

$$= T_l^{-1} y_d + T_r^{-1} \eta_{\cancel{ref}}$$

$$\dot{\eta} = T_\eta \dot{x} = T_\eta \left[A x + B u_{inv} \right]$$

$$= T_\eta \left[A x + B\left(\left[-B_y^{-1} A_y T_r^{-1} \right] \eta + \left[-B_y^{-1} A_y T_l^{-1} \;\vdots\; B_y^{-1} \right] \mathcal{Y}_d \right) \right]$$

$$= T_\eta \left[A\left(T_l^{-1} y_d + T_r^{-1} \eta \right) + B\left(-B_y^{-1} A_y T_r^{-1} \eta + \left[-B_y^{-1} A_y T_l^{-1} \mid B_y^{-1} \right] \begin{bmatrix} y_d \\ y_d^{(r)} \end{bmatrix} \right) \right]$$

$$= \underbrace{T_\eta \left[A T_r^{-1} - B B_y^{-1} A_y T_r^{-1} \right] \eta}_{A_{inv}\eta} + \underbrace{T_\eta \left[A T_l^{-1} - B B_y^{-1} A_y T_l^{-1} \mid B B_y^{-1} \right]}_{B_{inv}} \underbrace{\begin{bmatrix} y_d \\ y_d^{(r)} \end{bmatrix}}_{\mathcal{Y}_d}$$

Step 4: Write the Internal Dynamics

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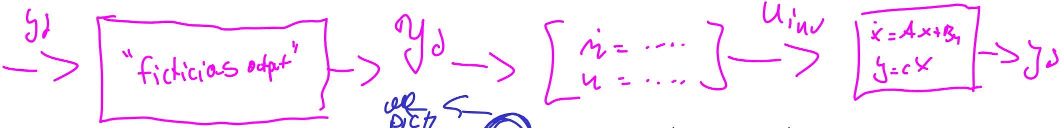
Step 5: Complete the Inverse System



we pick

needs to be stable!

$$A_{inv} = T_\eta \left[A - (BB_y^{-1}A_y)\right] T_r^{-1}$$

$$B_{inv} = \left[T_\eta \left[A - (BB_y^{-1}A_y)\right] T_l^{-1} \;\vdots\; T_\eta B B_y^{-1} \right]$$

$$C_{inv} = -B_y^{-1}A_y T_r^{-1}$$

$$D_{inv} = \left[-B_y^{-1}A_y T_l^{-1} \;\vdots\; B_y^{-1} \right]$$

$$\mathcal{Y}_d(t) = \begin{bmatrix} \xi_d(t) \\ y_d^{(r)}(t) \end{bmatrix}$$

$$\dot{\eta}_{ref} = A_{inv}\eta_{ref} + B_{inv}\mathcal{Y}_d$$

$$u_{inv} = C_{inv}\eta_{ref} + D_{inv}\mathcal{Y}_d$$

we observed this in simple example!

A_{inv}: evals independent of T_η

↳ eigenvalues are the zeros of the original system!

→ A_{inv} is unstable for non-min phase systems

Step 5.5: Solve the Inverse System

$$\dot{\eta} = A_{inv}\eta + B_{inv}y_d$$
$$u = C_{inv}\eta + D_{inv}y_d$$

we need $\eta(t_0)$

→ assume the system starts at equilibrium → we know!

$$\dot{x} = 0 \Rightarrow \dot{\eta} = 0 = A_{inv}\eta(t_0) + B_{inv}y_d(t_0)$$

$$\boxed{\eta(t_0) = -A_{inv}^{-1}B_{inv}y_d(t_0)}$$

↳ simulate for however long $y_d(t)$ is (say T)

$t_0 \to T \to u_{ff}$!

Step 6: For non-minimum phase systems

→ non-min phase zeros of original system → LHP eigenvalues of inv system

→ define T that splits η into its stable (RHP) evals and unstable (LHP) evals

$$T_{split}\begin{bmatrix}\eta_s\\ \eta_u\end{bmatrix} = \eta \rightarrow \text{matlab will do this!}$$

$$\frac{d}{dt}\begin{bmatrix}\eta_s\\ \eta_u\end{bmatrix} = T_{split}^{-1} A_{inv} T_{split}\begin{bmatrix}\eta_s\\ \eta_u\end{bmatrix} + T_{split}^{-1} B_{inv}\, y_d$$

(assume distinct evals)

$$= \begin{bmatrix}A_s & 0\\ 0 & A_u\end{bmatrix}\begin{bmatrix}\eta_s\\ \eta_u\end{bmatrix} + \begin{bmatrix}B_s\\ B_u\end{bmatrix} y_d$$

Step 6: For non-minimum phase systems

stable

$$\dot{\eta}_s = A_s \eta_s + B_s y_d$$

↳ solve forward in time

$$\eta_{s,ref}(t) = \int_{-\infty}^{t} e^{A_s(t-\tau)} B_s y_d(\tau)\, d\tau$$

initial cond

unstable

$$\dot{\eta}_u = A_u \eta_u + B_u y_d$$

↳ stable if you go backwards $t \to -t$

$e^{t} \to$ unstable $\Rightarrow e^{-t} \to$ stable

$$\dot{\eta}_u = -A_u \eta_u - B_u y_d$$

flipped

$$\eta_{u,ref}(t) = \int_{t}^{\infty} e^{-A_u(\tau - t)} B_u y(\tau)\, d\tau$$

$$\boxed{u_{inv}(t)} = C_{inv} \eta_{ref} + D_{inv} y_d \longleftarrow \eta_{ref} = T_{split} \begin{bmatrix} \eta_{s,ref} \\ \eta_{u,ref} \end{bmatrix}$$

Step 6: For non-minimum phase systems

Matlab Example

Non-Minimum Phase System

not going to: plug in the formulas

inverse system

y_d

nmp

stab/unstab

$$\begin{aligned} \dot{x} &= Ax + Bu_{ff} \\ y_d &= Cx \end{aligned} \quad \longrightarrow \quad \begin{aligned} \dot{\eta}_{ref} &= A_{inv}\eta_{ref} + B_{inv}\mathcal{Y}_d \\ u_{inv} &= C_{inv}\eta_{ref} + D_{inv}\mathcal{Y}_d \end{aligned} \quad \longrightarrow \quad \begin{aligned} \dot{\eta}_s &= A_s\eta_s + B_s\mathcal{Y}_d \\ \dot{\eta}_u &= A_u\eta_u + B_u\mathcal{Y}_d \end{aligned}$$

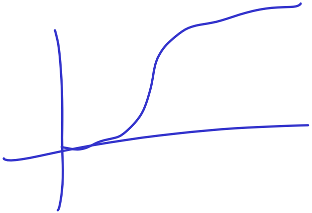
$\hookrightarrow u_{ff}$

fw

bw

η

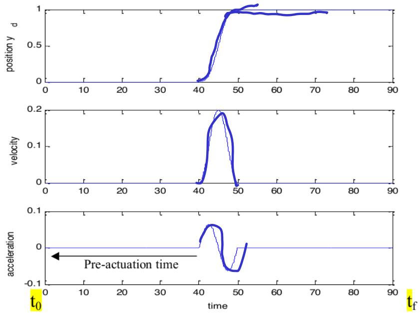
Focus: Internal Dynamics



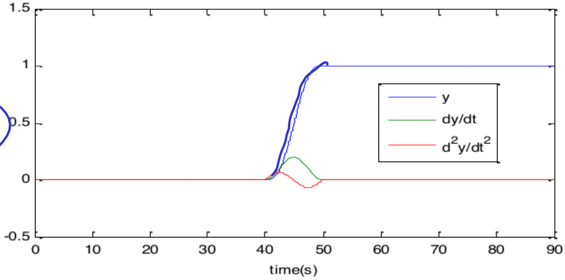
$$\dot{\eta}_s = A_s \eta_s + B_s \mathcal{Y}_d$$
$$\dot{\eta}_{\cancel{d}\,u} = A_u \eta_u + B_u \mathcal{Y}_d$$

$$\underline{A_s = -0.95}$$
$$\underline{A_u = 1.05}$$

Desired Trajectory



y_d



Solve Stable Subsystem

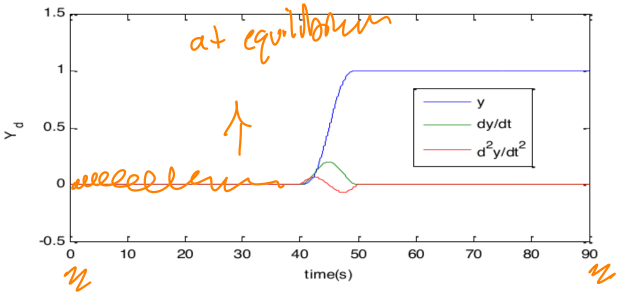
"output" just for Matlab

$$\dot{\eta}_s = A_s \eta_s + B_s \mathcal{Y}_d$$

$$Y_s = C_s \eta_s + D_s \mathcal{Y}_d$$

$$C_s = \begin{bmatrix} 1 \end{bmatrix}, \quad D_s = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

$$\mathcal{Y}_d = \begin{bmatrix} y & \dot{y} & \ddot{y} \end{bmatrix}^T$$



Solve Stable Subsystem

$\dot{x} = Ax + Bu$

$y = Cx + Du$

lsim $\rightarrow$ returns both x and y

η

$$\dot{\eta}_s = A_s \eta_s + B_s \mathcal{Y}_d$$

$$Y_s = C_s \eta_s + \cancel{D_s \mathcal{Y}_d}$$

$$C_s = [1], \; D_s = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

$$\mathcal{Y}_d = \begin{bmatrix} y & \dot{y} & \ddot{y} \end{bmatrix}^T$$

Initial condition (from equilibrium:)

statespace

$$0 = A_s \eta_s + B_s \mathcal{Y}_d$$

$$\eta_s(t_0) = -A_s^{-1} B_s \mathcal{Y}_d(t_0)$$

(nothing to do with methods)

vector of integers

sys_s=ss(A_s, B_s, C_s, D_s)

[Y_s, t, $\eta_{s,ref}$]=lsim(sys_s, $\mathcal{Y}_d$, t, $\eta_s(t_0)$)

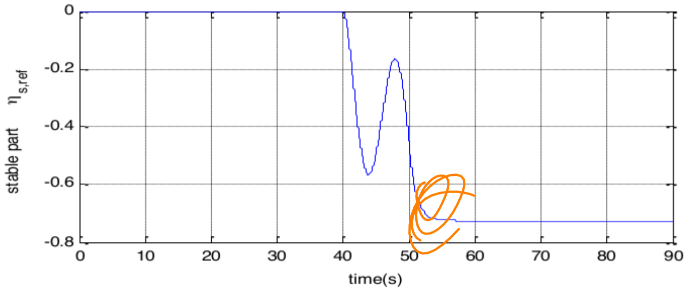
timespan

linear system simulator

$C_s \eta_s$

$= 1 \cdot \eta_s$

Solve Stable Subsystem



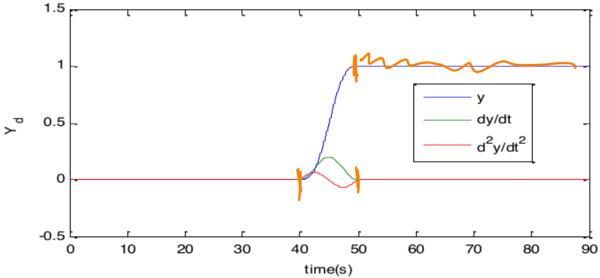
Solve Unstable Subsystem

$$\dot{\eta}_u = A_u \eta_u + B_u \mathcal{Y}_d$$

$$Y_u = \underline{C_u \eta_u} + \cancel{D_u \mathcal{Y}_d}$$

$$C_u = \begin{bmatrix} 1 \end{bmatrix}, \; D_u = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

$$\mathcal{Y}_d = \begin{bmatrix} y & \dot{y} & \ddot{y} \end{bmatrix}^T$$



Solve Unstable Subsystem

$$\dot{\eta}_u = A_u \eta_u + B_u \mathcal{Y}_d$$
$$Y_u = C_u \eta_u + D_u \mathcal{Y}_d$$
$$C_u = \begin{bmatrix} 1 \end{bmatrix}, \ D_u = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$
$$\mathcal{Y}_d = \begin{bmatrix} y & \dot{y} & \ddot{y} \end{bmatrix}^T$$

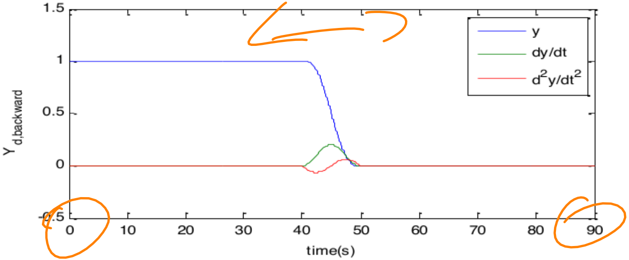
Solve it backwards in time:

$$\dot{\eta}_u = -A_u \eta_u + -B_u \mathcal{Y}_d$$
$$Y_u = C_u \eta_u + D_u \mathcal{Y}_d$$

Solve Unstable Subsystem

$$\dot{\eta}_u = A_u\eta_u + B_u\mathcal{Y}_d$$
$$Y_u = C_u\eta_u + D_u\mathcal{Y}_d$$
$$C_u = \begin{bmatrix}1\end{bmatrix}, \; D_u = \begin{bmatrix}0 & 0 & 0\end{bmatrix}$$
$$\mathcal{Y}_d = \begin{bmatrix}y & \dot{y} & \ddot{y}\end{bmatrix}^T$$

Solve it backwards in time:



Solve Unstable Subsystem

$$\dot{\eta}_u = -A_u\eta_u + -B_u\mathcal{Y}_d$$
$$Y_u = C_u\eta_u + D_u\mathcal{Y}_d$$

Initial condition (from equilibrium:)

90 = tf

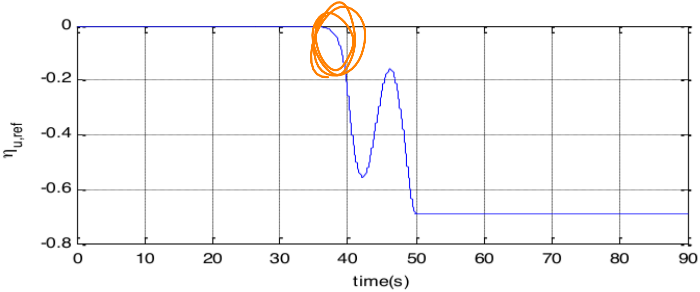
$$0 = A_u\eta_u + B_u\mathcal{Y}_d$$
$$\eta_u(t_f) = -A_u^{-1}B_u\mathcal{Y}_d(t_f)$$

time t

0 90

- sys_u_bw = ss($-A_u$, $-B_u$, C_u , D_u)
- $\mathcal{Y}_{d.bw}$ = fliplr($\mathcal{Y}_d$)
- [Y_u , t, η_{ub}] = lsim(sys_u_bw, $\mathcal{Y}_{d,bw}$, $\eta_u(t_f)$)
- $\eta_{u,ref}$ = fliplr(η_u^T)

Solve un Stable Subsystem



Find Reference Input

lsim (ss(A,B,C,D), uff, t, x(t0))

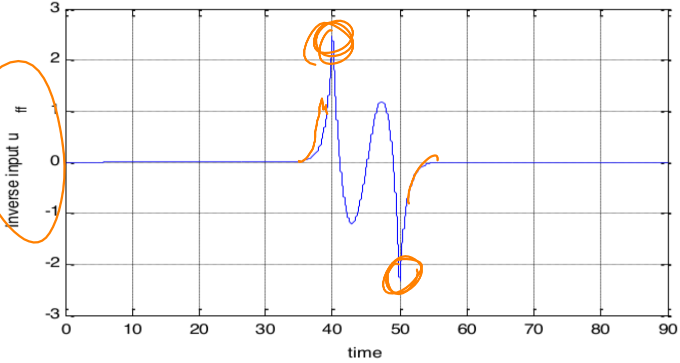
↳ yd

η_{ref}

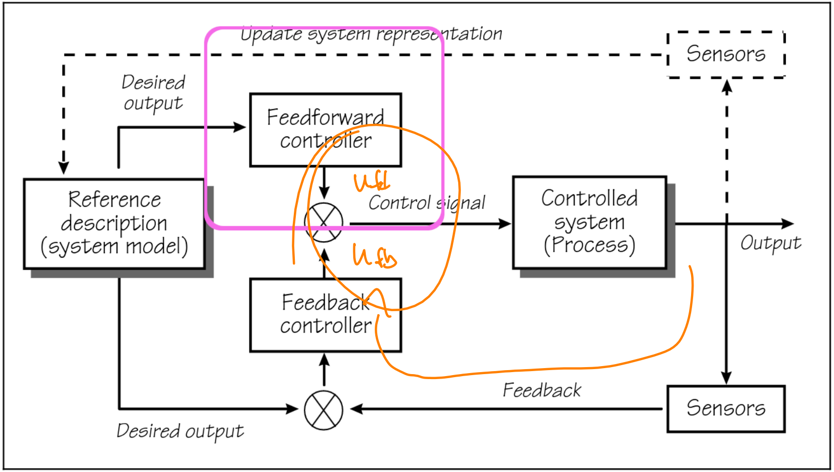
$$\eta_{ref} = T_{split} \begin{bmatrix} \eta_{s,ref} \\ \eta_{u,ref} \end{bmatrix}$$

$$u_{inv}(t) = C_{inv}\eta_{ref} + D_{inv}\mathcal{Y}_d$$

↳



Today



Update system representation
Sensors
Desired output
Feedforward controller
u_{ff}
Control signal
Reference description (system model)
Controlled system (Process)
Output
u_{fb}
Feedback controller
Feedback
Sensors
Desired output

Next Time

system identification

↪ $G(s)$ from $\{u_k, y_k\}$